

make decisions and manage funds autonomously, based on the rules defined in their smart contracts.

Consensus mechanism: Ethereum currently operates on a proof of stake (PoS) consensus mechanism known as Ethereum 2.0 (Eth2). This is an upgrade from the original proof of work (PoW) mechanism, which is still used in Ethereum 1.0. Eth2 aims to improve scalability, security, and energy efficiency.

EVM and Solidity: Ethereum introduced the Ethereum Virtual Machine (EVM), a runtime environment that enables smart contracts to be executed. Solidity is the most commonly used programming language for writing smart contracts on Ethereum.

Interoperability: Ethereum has supported the development of various standards and protocols, such as ERC-20 (fungible tokens) and ERC-721 (non-fungible tokens), which have contributed to the growth of the blockchain ecosystem.

Upgrades: Ethereum is continually evolving. Major upgrades are being rolled out to address scalability, security, and sustainability challenges.

It's important to note that Ethereum, like other blockchains, requires participants (nodes) to validate transactions and maintain the network. This decentralised nature helps ensure security, censorship resistance, and immutability of the blockchain's data.

In order to work with Ethereum on AWS, we may consider using Amazon Managed Ethereum (AME) or deploying our Ethereum nodes manually on AWS services such as EC2 instances or Kubernetes clusters.

Amazon Managed Ethereum (AME)

AME is a fully managed Ethereum service provided by AWS. It is designed to simplify the deployment and management of Ethereum infrastructure.

It allows the creation of Ethereum networks quickly without the need to manage the underlying infrastructure. We can use AME to deploy Ethereum nodes, interact with smart contracts, and build decentralised applications (DApps) on the Ethereum network.

Solution for land grabbing prevention

Blockchain technology, when used in conjunction with legal and societal reforms, can help mitigate land grabbing by increasing transparency, improving property record management, and enhancing trust. Let us see how Ethereum blockchain technology can contribute to the prevention of land grabbing.

Immutable land records:

Ethereum's blockchain offers immutability, meaning that once data is recorded on the blockchain, it is extremely difficult to alter or delete it. Land records stored on Ethereum can provide a tamper-resistant history of ownership and transactions, reducing the risk of fraudulent changes to land titles.

Transparent land registries: Using Ethereum-based smart contracts, governments or organisations can create transparent land registries that are accessible to the public.

Identity verification: Implementing a blockchain-based identity verification system can ensure that only legitimate property owners or authorised parties can initiate land transactions. This can reduce the chances of unauthorised land sales.

Smart contracts for transactions:

Ethereum's smart contracts can be used to automate and validate land transactions. When a transaction is initiated, the smart contract can verify the legitimacy of the buyer and seller, and ensure that all legal requirements are met before transferring ownership.

Timestamping and notarisation:

Blockchain can be used for timestamping and notarisation of land-related documents, creating a provable history of documents related to land transactions.

Community involvement:

Blockchain can facilitate community involvement and consensus in land management decisions. Community members can collectively validate and approve land transactions to prevent unilateral decisions that may lead to land grabbing.

Legal and regulatory frameworks:

While blockchain can provide technical solutions, it must be integrated into a supportive legal and regulatory framework. Governments and authorities should recognise blockchain-based land records and transactions as legally binding.

It's important to recognise that blockchain technology is not a silver bullet for preventing land grabbing. It should be part of a broader strategy that includes legal reforms, community engagement, and good governance practices. Additionally, blockchain systems should be designed with careful consideration of privacy, security, and scalability to be effective in land management. **END** 

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